

SAMUEL MWANGI NGONJO

Full-Stack Developer & Systems Engineer

Phone: 0799594621

Email: samuelmwas262@gmail.com

LinkedIn: <https://linkedin.com/in/samuel-mwangi-11597a245>

GitHub: <https://github.com/scriptedSyntax>

For a detailed overview of my work and additional projects, please visit my portfolio at

<https://scriptedsyntax.github.io/portfolio/>

Professional Summary

Full-stack developer and systems engineer specialising in backend systems, AI systems, frontend frameworks, and hardware–software integrations. Experienced in building and scaling Django and React-based applications, automating operational workflows, and integrating AI services for intelligent search, document processing, and automation. With a biomedical engineering foundation, I build reliable and maintainable systems for healthcare and operational environments.

For examples of my work, including live projects and technical demonstrations, please refer to my portfolio.

Core Skills

- **Backend Engineering & Integrations**
 - Django, Flask, FastAPI, Python, RESTful APIs
 - PostgreSQL, SQLite, Firebase, MySQL
 - Authentication, security, data interoperability
 - Cloud-based deployment & system scaling
- **Frontend & Full-Stack**
 - React, Next.js, CSS, HTML
 - Integration of AI models and workflows into user-facing applications
 - Hardware–software user interfaces for connected devices
- **Automation & Operational Efficiency**
 - Workflow automation & system optimisation
 - Health tech system digitization
 - Logging & monitoring
 - API-driven process improvement
- **Data & Infrastructure**
 - Scripting, and reporting
 - Pandas, NumPy, analytics workflows
 - IoT integrations
 - Linux, Git
- **Tech Leadership & Enablement**
 - Technical training & team mentoring
 - Communication across clinical & technical teams
 - Requirements gathering & solution architecture
 - Documentation, SOP development, and compliance support

Experience

Senior Software Developer / Full-Stack Developer – GONEP Pharmaceuticals | Aug 2025 – Dec 2025

- Developed and deployed scalable internal applications across departments, working end-to-end from database schema to user interface.
- Modernised utility and lab systems, improving reliability and efficiency and reducing manual effort.
- Built the clinic-at-hand lab system integrating hardware and software for point-of-care workflows (<https://gonepharm.com>).
- Migrated legacy systems to modern platforms and automated compliance reporting, cutting manual effort by 60%+.
- Collaborated with cross-functional teams to optimise system performance and reduce downtime by over 70%.
- Implemented CI/CD pipelines and cloud-ready architecture for rapid, reliable releases.

Backend & AI Engineer / Full-Stack Developer – OnlineMedic Healthcare | Nov 2024 – Mar 2025

- Designed and deployed Python (Django) backend services supporting AI-enabled healthcare workflows.
- Integrated AI modules for document understanding, data extraction, and automation-ready pipelines.
- Built scalable REST APIs and modern React-based interfaces for AI-driven features across web and mobile platforms.

Independent & Freelance Full-Stack Developer | 2022 – Present

- Delivered full-stack web applications for small businesses and startups, handling design, development, and deployment.
- Developed responsive websites such as Ledanie Construction (<https://ledanieconstruction.com>) and GONEP Pharma (<https://gonepharm.com>).
- Implemented user-friendly interfaces and robust backend architectures, including authentication, data handling and API integrations.
- Provided ongoing support and enhancements, adapting to customer feedback and evolving business requirements.

Biomedical Engineering Technologist – Kenyatta University Teaching, Referral & Research Hospital – Level 6 | Mar 2024 – May 2024

- Maintained and repaired medical equipment, ensuring continuous clinical operation.
- Diagnosed and resolved breakdowns promptly to minimise downtime.
- Deployed and tested new machines, training staff for smooth adoption.
- Developed SOPs and digital tracking systems to streamline workflows and monitor devices.

Biomedical Engineering Technologist – Thika Level 5 Hospital | Nov 2023 – Jan 2024

- Modernised utility and imaging systems to improve reliability.
- Optimised preventive maintenance, reducing downtime by 20%.
- Implemented a digital CMMS for efficient equipment tracking.
- Ensured compliance through regular inspections and documentation.

Education

Diploma in Medical Engineering Technology – Kenya Medical Training College – 2024

- Key subjects: Clinical Engineering, Health Informatics, Medical Device Design, Engineering Math, Mechanical Systems, Electronics.

Computer Science in Healthcare (Self-Driven Education)

- Learned through various platforms and improved my skills through continuous learning and development. This was accompanied by hands-on projects that contributed to my skills.
- **Technical Focus Areas:** Python backend development, Django, Flask & FastAPI APIs, relational and NoSQL databases, AI-enabled backend systems, document processing and intelligent search workflows, RAG patterns, cloud infrastructure, CI/CD-ready deployment, cybersecurity and data privacy compliance, healthcare systems interoperability, workflow automation, and medical device interface programming.

Certifications & Training

- Medical Device QA & Safety – Defibrillators, NIBP, Neonatal Devices, Electrical Safety (Fluke Biomedical)
- Advanced Imaging QA – PET/CT & SPECT/CT (GE Healthcare)
- Clinical Systems & Simulation – Oxygen QA, ProSim 4 & 8 (Fluke Biomedical)

Skills

Technical Skills

- Programming & Development
- Web & Backend
- Tools & Workflow
- Systems & IoT Integration
- Operational & Compliance Systems
- GitHub Actions
- Cloud deployment

Soft Skills

- Clear communicator across technical and non-technical teams
- Reliable in teams or independent roles
- Strong problem-solving
- Critical thinking
- Field-tested in hands-on operations
- Adapts fast to new technologies and workflows
- Effective collaborator
- Technical documentation and reporting
- Client communication and on-site support
- Team coordination and field leadership

References

Available upon request.